CSE220 Digital Audio Equalizer -- DSP Analysis Report

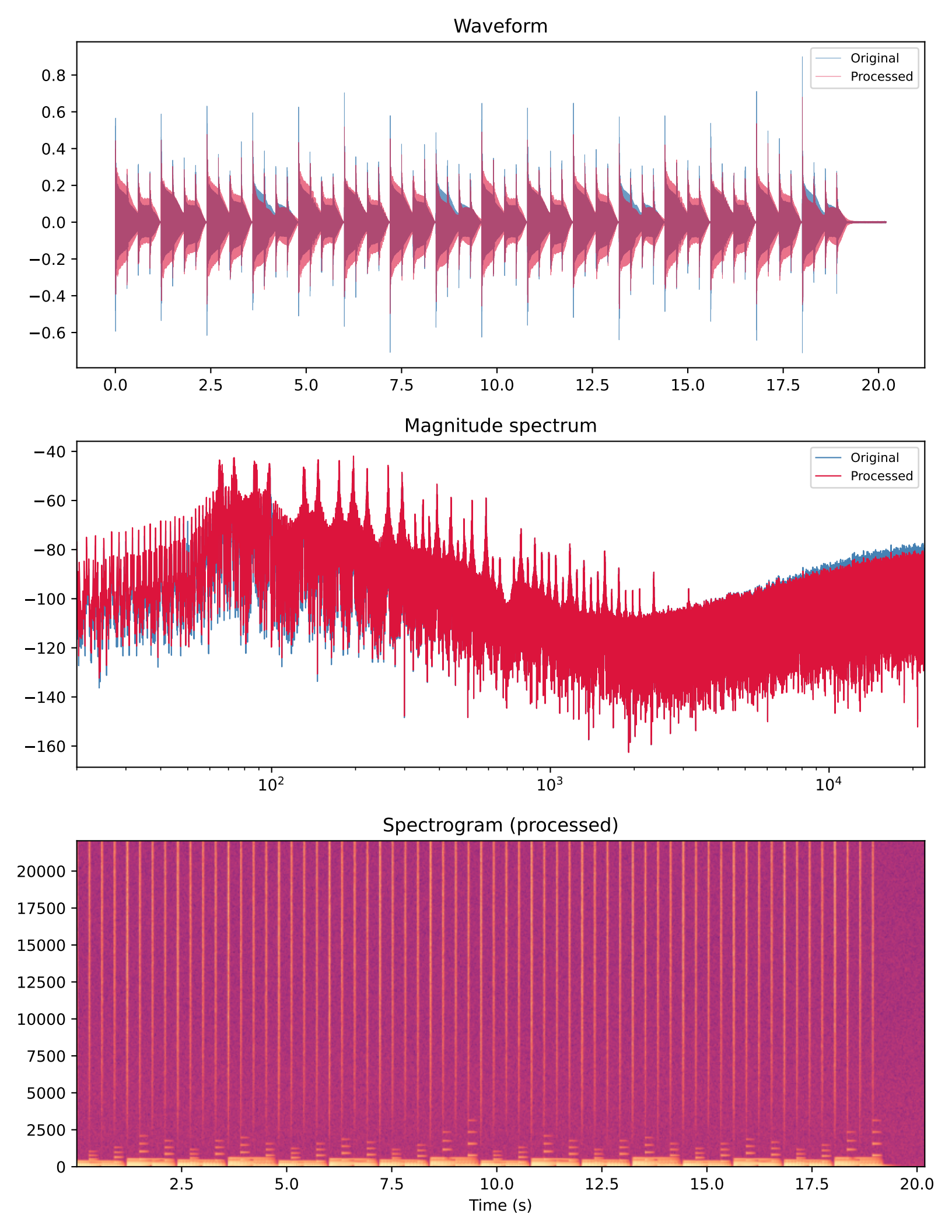
```
Mode: parametric
EQ settings: parametric preset 'warm' bands=[{'type': 'lowshelf', 'f0': 200, 'gain_db': 4, 'Q'

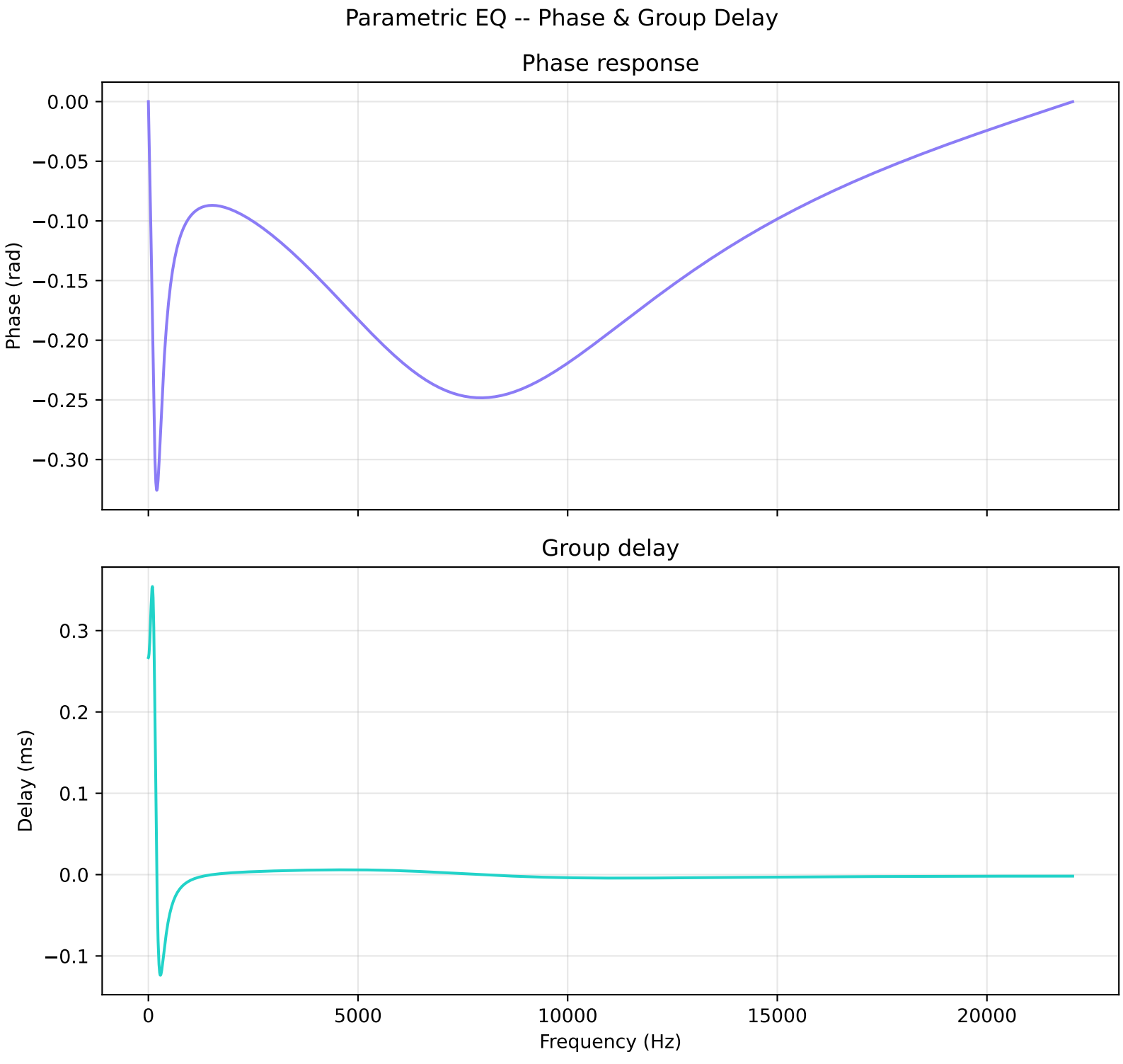
Sample rate: 44100 Hz
Duration: 20.20 s

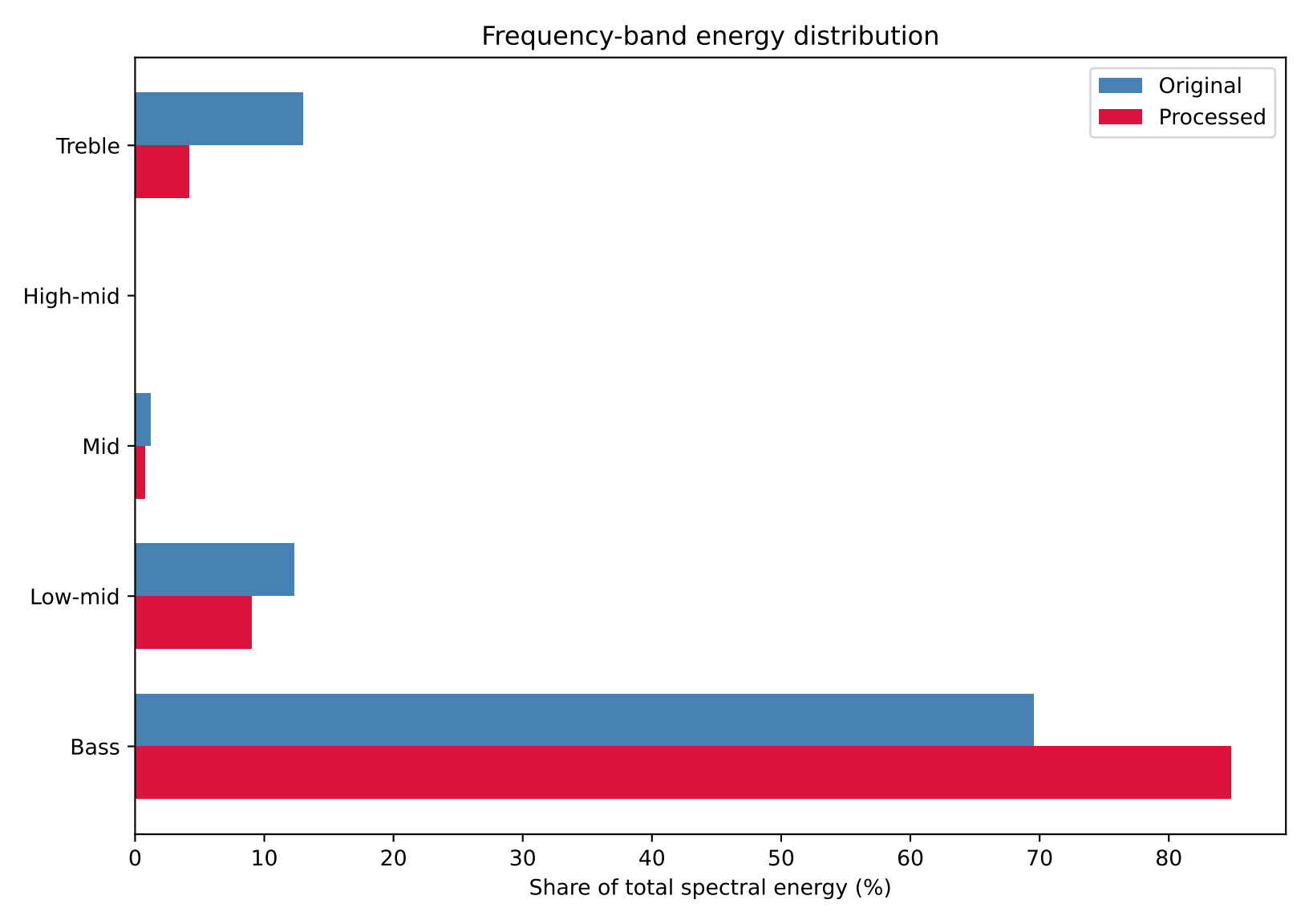
Clipping protection:
  Peak before normalization : 0.6794
  Peak after normalization  : 0.6794
  Clipping detected         : False

Quantitative metrics (Original vs Processed):

Metric                         Original     Processed
-----------------------------------------------------
Duration (s)                      20.20         20.20
RMS                              0.0600        0.0766
Peak                             0.9000        0.6794
Energy                         3210.88       5229.63
Spectral Centroid (Hz)         13042.3       11553.7
Dominant Freq (Hz)               196.7         196.7
Zero-Crossing Rate              0.1039        0.0671
```


Waveform
Original
Processed
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Magnitude spectrum
Original
Processed
−40
−60
−80
−100
−120
−140
−160
10^2
10^3
10^4
Spectrogram (processed)
20000
17500
15000
12500
10000
7500
5000
2500
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Parametric EQ -- Phase & Group Delay
Phase response
Phase (rad)
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
Group delay
Delay (ms)
0.3
0.2
0.1
0.0
−0.1
0
5000
10000
15000
20000
Frequency (Hz)


Frequency-band energy distribution
Original
Processed
Treble
High-mid
Mid
Low-mid
Bass
0
10
20
30
40
50
60
70
80
Share of total spectral energy (%)

Filter Performance & FIR vs IIR

```
Filter performance (design target vs measured):
  Target frequency   : 200.0 Hz
  Target gain        : +4.00 dB
  Measured gain      : +2.02 dB
  Gain error         : -1.9842 dB
  Measured peak freq : 0.0 Hz
  -3dB bandwidth     : 269.2 Hz

FIR vs IIR comparison:

Property         FIR (Graphic EQ)                          IIR (Parametric EQ)
--------------------------------------------------------------------------------
Structure        Parallel bank of 10 band-pass FIR filters  Cascade of 2 biquad section(s)
Stability        Inherently stable (no poles)               Stable (max |pole|=0.9820)
Phase            Linear (symmetric kernel)                  Nonlinear (typical for IIR)
Coefficients     401                                        10
Processing time  599.01 ms                                  13.06 ms
```